EXAM PAPERS PRACTICE

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AQA A Level Further Mathematics (7367)

Topic D: Further Algebra and Functions

Sub topic D16: Transformations of Graphs

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic D: Further Algebra and Functions

Sub topic D16: Transformations of Graphs

To be used by all students and teachers for AQA A Level Further Mathematics (7367) for exams from 2025 onwards.

Students and teachers of other boards may also find this useful

Name : ______________________

Date : ______________________

Time : ______________________

Total Marks Available : _____

Total Marks Received : _____

Q1.

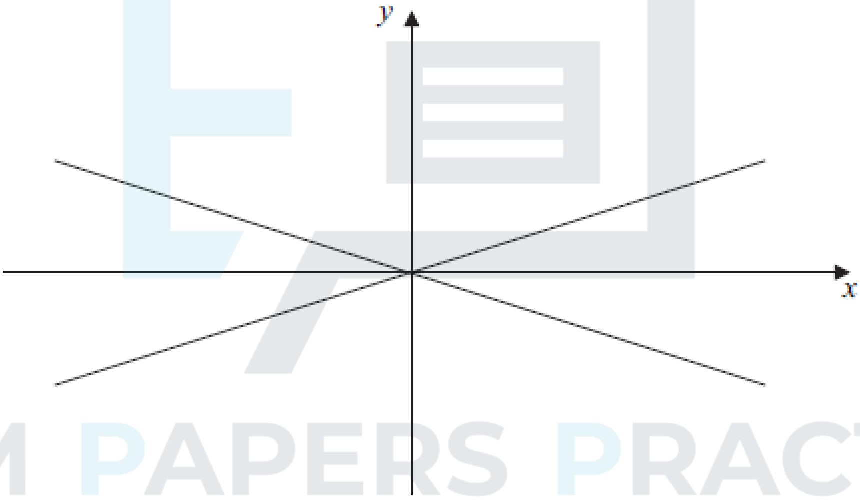
A hyperbola H has equation

$$\frac{x^2}{9} - y^2 = 1$$

(a) Find the equations of the asymptotes of H.

(1)

(b) The asymptotes of H are shown in the diagram below. On the same diagram, sketch the hyperbola H. Indicate on your sketch the coordinates of the points of intersection of H with the coordinate axes.



(3)

(c) The hyperbola H is now translated by the vector $\begin{bmatrix} -3 \\ 0 \end{bmatrix}$.

(i) Write down the equation of the translated curve.

(2)

(ii) Calculate the coordinates of the two points of intersection of the translated curve with the line $y = x$.

(4)

(d) From your answers to part (c)(ii), **deduce** the coordinates of the points of intersection of the original hyperbola H with the line $y = x - 3$.

(2)

(Total 12 marks)

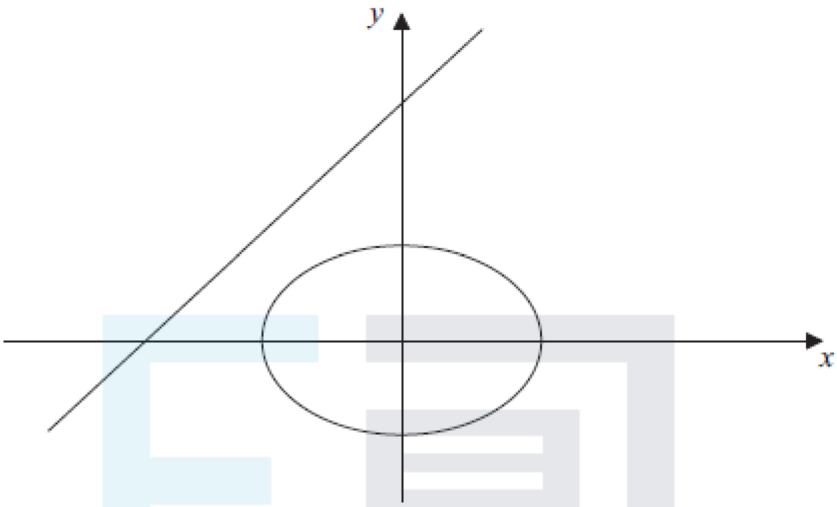
Q2.

The diagram shows the ellipse E with equation

$$\frac{x^2}{5} + \frac{y^2}{4} = 1$$

and the straight line L with equation

$$y = x + 4$$



(a) Write down the coordinates of the points where the ellipse E intersects the coordinate axes.

(2)

(b) The ellipse E is translated by the vector $\begin{bmatrix} p \\ 0 \end{bmatrix}$, where p is a constant. Write down the equation of the translated ellipse.

(2)

(c) Show that, if the translated ellipse intersects the line L, the x-coordinates of the points of intersection must satisfy the equation

$$9x^2 - (8p - 40)x + (4p^2 + 60) = 0$$

(3)

(d) Given that the line L is a tangent to the translated ellipse, find the coordinates of the two possible points of contact.

(No credit will be given for solutions based on differentiation.)

(8)

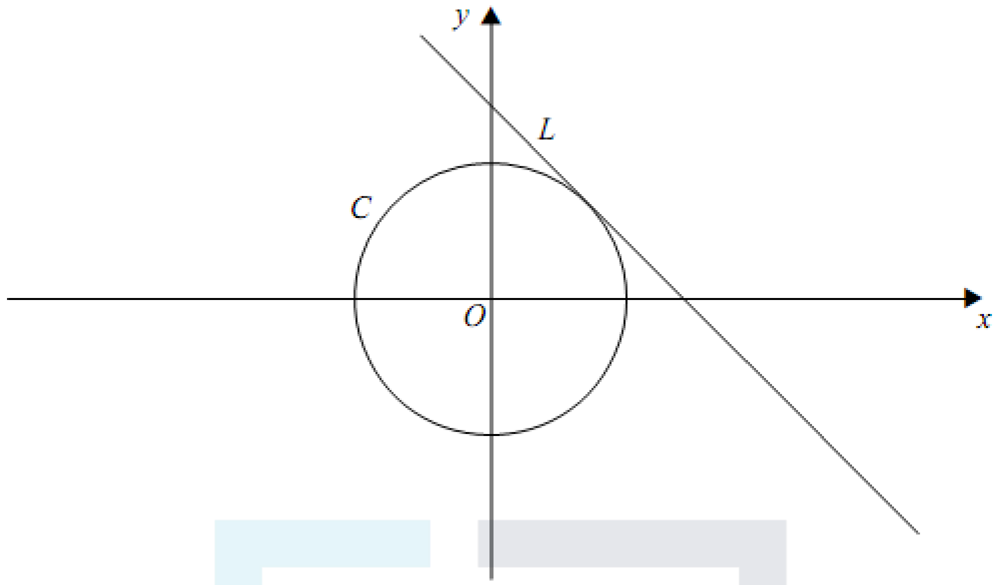
(Total 15 marks)

Q3.

The diagram shows a circle C and a line L, which is the tangent to C at the point (1, 1). The equations of C and L are

$$x^2 + y^2 = 2 \text{ and } x + y = 2$$

respectively.



The circle C is now transformed by a stretch with scale factor 2 parallel to the x-axis. The image of C under this stretch is an ellipse E.

(a) **On the diagram below,** sketch the ellipse E, indicating the coordinates of the points where it intersects the coordinate axes.

(4)

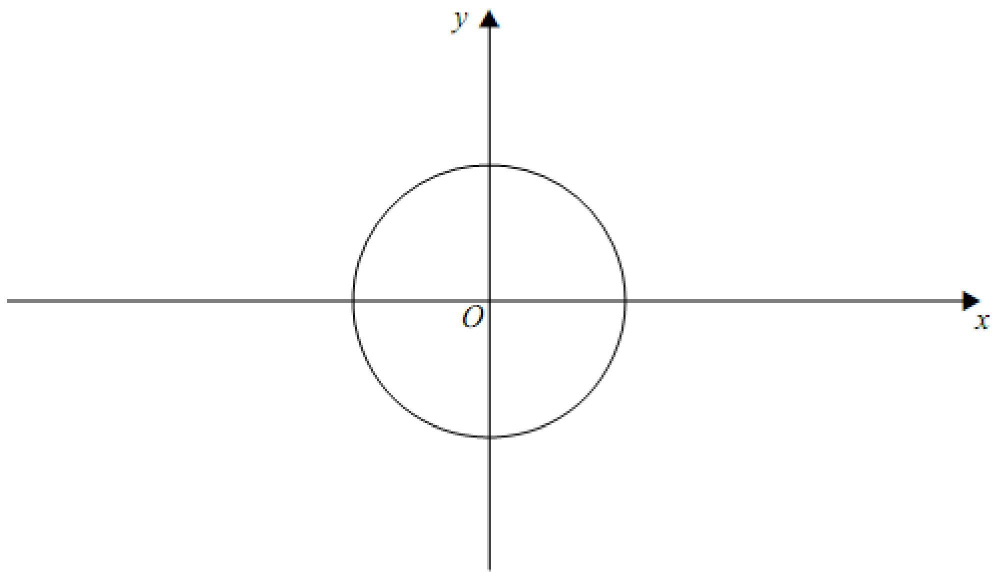
(b) Find equations of:

(i) the ellipse E;

(2)

(ii) the tangent to E at the point (2, 1).

(2)



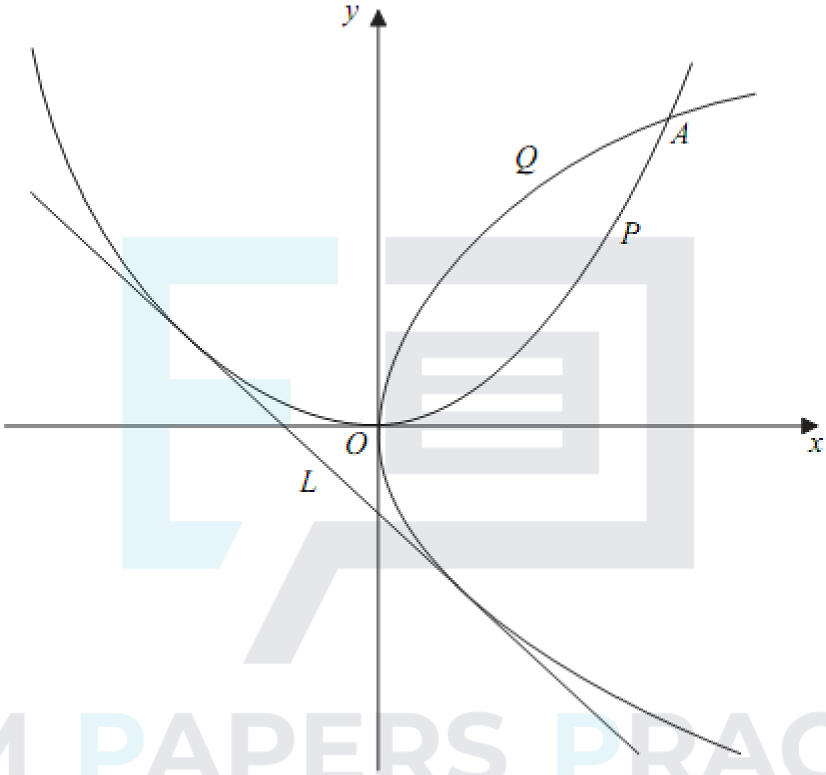
(Total 8 marks)

Q4.

The diagram shows a parabola P which has equation $y = \frac{1}{8}x^2$, and another parabola. Q which is the image of P under a reflection in the line $y = x$.

The parabolas P and Q intersect at the origin and again at a point A.

The line L is a tangent to both P and Q.



(a) (i) Find the coordinates of the point A.

(2)

(ii) Write down an equation for Q.

(1)

(iii) Give a reason why the gradient of L must be –1

(1)

(b) (i) Given that the line $y = -x + c$ intersects the parabola P at two distinct points, show that

$$c > -2$$

(3)

(ii) Find the coordinates of the points at which the line L touches the parabolas P and Q.

(No credit will be given for solutions based on differentiation.)

(4)

(Total 11 marks)

Q5.

An ellipse E has equation

$$\frac{x^2}{3}+\frac{y^2}{4}=1$$

(a) Sketch the ellipse E, showing the coordinates of the points of intersection of the ellipse with the coordinate axes.

(3)

(b) The ellipse E is stretched with scale factor 2 parallel to the y-axis.

Find and simplify the equation of the curve after the stretch.

(3)

(c) The **original** ellipse, E, is translated by the vector $\begin{bmatrix} a \\ b \end{bmatrix}$. The equation of the translated ellipse is

$$4x^2 + 3y^2 - 8x + 6y = 5$$

Find the values of a and b.

(5)

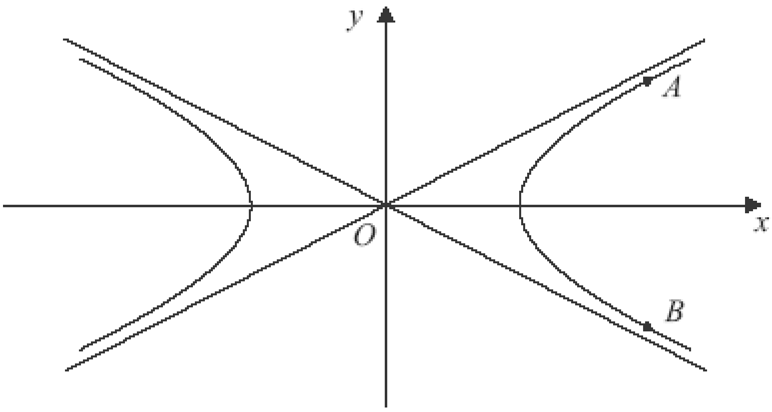
(Total 11 marks)

Q6.

The diagram shows the hyperbola

$$\frac{x^2}{4}-y^2=1$$

and its asymptotes.



(a) Write down the equations of the two asymptotes.

(2)

(b) The points on the hyperbola for which $x = 4$ are denoted by A and B.

Find, in surd form, the y-coordinates of A and B.

(2)

(c) The hyperbola and its asymptotes are translated by two units in the positive y direction.

Write down:

(i) the y-coordinates of the image points of A and B under this translation;

(1)

(ii) the equations of the hyperbola and the asymptotes after the translation.

(3)
(Total 8 marks)

Q7.

A curve C has equation

$$y = 7 + \frac{1}{x+1}$$

(a) Define the translation which transforms the curve with equation $y = \frac{1}{x}$ onto the curve C.

(2)

(b) (i) Write down the equations of the two asymptotes of C.

(2)

(ii) Find the coordinates of the points where the curve C intersects the coordinate axes.

(3)

(c) Sketch the curve C and its two asymptotes.

(3)
(Total 10 marks)

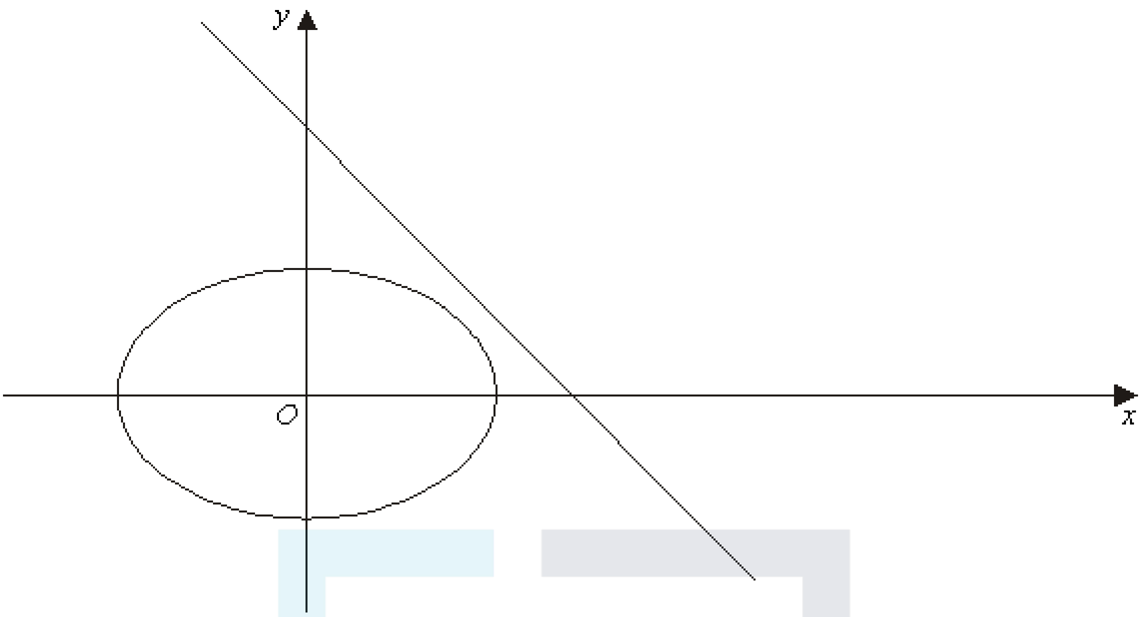
Q8.

The diagram below shows the curve with equation

$$\frac{x^2}{2} + y^2 = 1$$

and the straight line with equation

$x + y = 2$



(a) Write down the exact coordinates of the points where the curve with equation $\frac{x^2}{2} + y^2 = 1$ intersects the coordinate axes.

(2)

(b) The curve is translated k units in the positive x direction, where k is a constant. Write down, in terms of k, the equation of the curve after this translation.

(2)

(c) Show that, if the line $x + y = 2$ intersects the **translated** curve, the x-coordinates of the points of intersection must satisfy the equation

$$3x^2 - 2(k + 4)x + (k^2 + 6) = 0$$

(4)

(d) Hence find the two values of k for which the line $x + y = 2$ is a tangent to the translated curve. Give your answer in the form $p \pm \sqrt{q}$, where p and q are integers.

(4)

(e) On the figure, show the translated curves corresponding to these two values of k.

(3)

(Total 15 marks)

Q9.

A curve has equation $y^2 = 12x$.

(a) Sketch the curve.

(2)

(b) (i) The curve is translated by 2 units in the positive y direction. Write down the equation of the curve after this translation.

(2)

(ii) The **original** curve is reflected in the line $y = x$. Write down the equation of the curve after this reflection.

(1)

(c) (i) Show that if the straight line $y = x + c$, where c is a constant, intersects the curve $y^2 = 12x$, then the x-coordinates of the points of intersection satisfy the equation

$$x^2 + (2c - 12)x + c^2 = 0$$

(3)

(ii) Hence find the value of c for which the straight line is a tangent to the curve.

(2)

(iii) Using this value of c, find the coordinates of the point where the line touches the curve.

(2)

(iv) In the case where $c = 4$, determine whether the line intersects the curve or not.

(3)

(Total 15 marks)

Q10.

(a) Describe a geometrical transformation by which the hyperbola

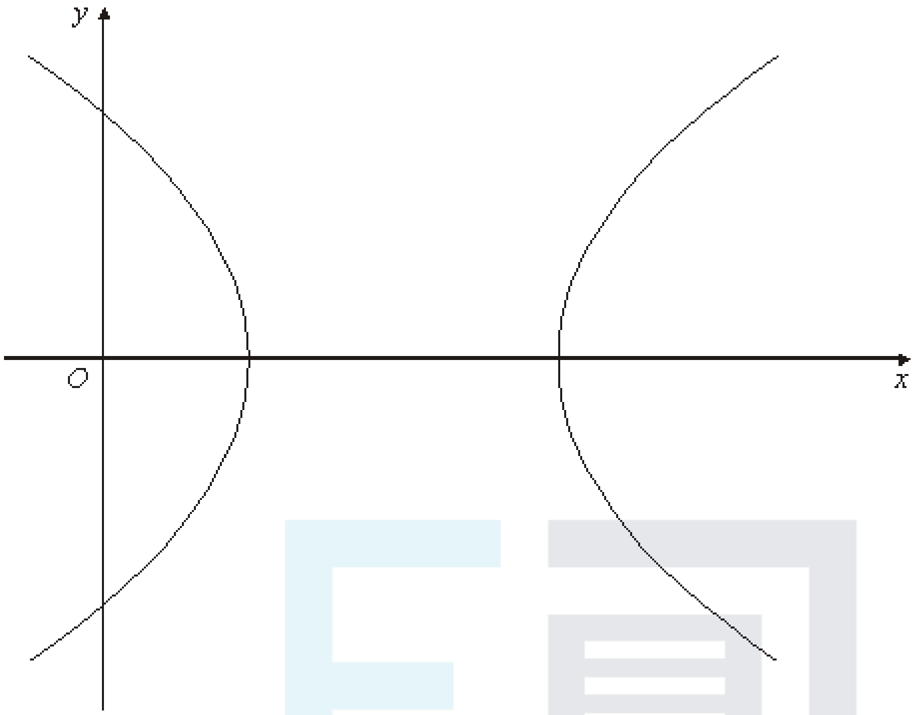
$$x^2 - 4y^2 = 1$$

can be obtained from hyperbola $x^2 - y^2 = 1$.

(2)

(b) The diagram shows the hyperbola H with equation

$$x^2 - y^2 - 4x + 3 = 0$$



By completing the square, describe a geometrical transformation by which the hyperbola H can be obtained from hyperbola $x^2 - y^2 = 1$.

(4)
(Total 6 marks)

Q11.

A curve has equation

$$\frac{x^2}{9} + \frac{y^2}{4} = 1$$

(a) Sketch the curve, showing the coordinates of the points of intersection with the coordinate axes.

(3)

(b) Calculate the y-coordinates of the points of intersection of the curve with the line $x = 1$. Give your answers in the form $p\sqrt{2}$, where p is a rational number.

(3)

(c) The curve is translated one unit in the positive x direction. Write down the equation of the curve after the translation.

(2)
(Total 8 marks)